

Amarm 50219 AgSp1.1 cr	PGNTPGAQLGPDGHINRIVLPTPA	PNPPSSGPLNPAGEP	.TAPFGPGNSPNSP	.QSPGNYPGSAFKFPPGFPGAPG	ANGPIGYIDFSQLPSGESPE	93
Lcorn 51676 AgSp1.1Cfrag cr	PGNTPGAELGPDGQITRIILPTPPP	PNPPSSGPLNPAGEP	.IAPFGPGNSPNSP	.QSPGNYPGSAFQFPFGFPGAPG	SNGPIGYLDIFSQLPSDVSP	93
Lcorn 58650 AgSp1.2 cr	PGNTPGAELGPDGQITRIILPTPPP	PNPPSSGPLNPAGEP	.IAPFGPGNSPNSP	.QSPGNYPGSAFQFPFGFPGAPG	SNGPIGYLDIFSQLPSDVSP	93
Ncruc 58532and13532 AgSp1 cr	PGNTPGAEMGPDGHINRIVLPTP	SPNPQSSGPQYPGGGGP	GGPFGPGNSPNSPG	.QSPGNYPGSAFQFPFGFPGAPG	SNGPIGYIDFSQLP	95
Mlaby 7048 AgSp1 cr	PGHTPGAELGPNGNINRIFLPTA	PNPPSPGPLNPPGEP	.IAPFGPGNSPNSP	.QSPGNYPGSAFQFPFGFPGAPG	SNGPIGYIDFSQLP	93
Mhut 1271 AgSp1 cr	PGITPGVELGPDGHINRIVLPTP	PPNPLFSDPLNPVKGK	.AAPFGPGNIPNSP	.YSPGNYPGSAFQFPFGFPGAPG	SNGPIGYIDFSQLPSGVSPE	93
Aarg AgSp1 FC511 cr	PGTTPGAELGPDGKINKIVLPTT	PPNPPSSGPLNPENEP	.IPFGPGNSPNSP	.QTPGNYPGSAFQFPFGPDAPG	SNGPIGYIDFSQLPSGESPE	93
Atri AgSp1 genomic cr	PGTTPGAELGPDGKINKIILPTT	PPNPPSSGPLNPKGEP	.IPFGPGNSPNSP	.QPPGNYPGSAFQFPFGPDAPG	ANGPIGYIDFSQLPSGESPE	93
Msag 19329 AgSp1.1 cr				PGTTPGVVTGSDG.....	QPVK.....Y	18
Msag 18528 AgSp1.2 cr	PGTTPGVRIDPDGKINNVLVLP	ISPNSSELGGLNPDNEP	.IPFGPDNTPNSP	.NFGPNYPRSTFKFPGFPGAPG	SMGPIGFLDFSELS..ESPE	91
Mg TRINITY DN1546 c0 g1 Cr	GGNTPGAELDPDGQIKKVILP	STPSEPFSSSSLPSSQ			DSQLNGREPYE	47
Varen 6869 AgSp1.1 cr	PGNNPGVEMGPDGQINKIILP	STPANPSSSGNPNNG..Q				36
Varen 6869 AgSp1.2 cr	PGNTPGAELGPDGQINRIILP	STPPNPPSSGPLNPDGEP	.SAPFGPGNSFNSPK.....	SPGFKFPGFPGAPGSMGPVGM	DFSQLPSGESPE	87

Amarm 50219 AgSp1.1 cr	MVGSIGFFPNFSPEIGGPFPGFS	SGGPGNSGPGGFLNINSLP	DFMNPVYGFPGSPQAPLG	YLDKSM	LPPGFNLDS	SSGQLVFP	PGDPGSPGSGSQFP	188	
Lcorn 51676 AgSp1.1Cfrag cr	MVGPIGFFPNFNPEIGGQFP	GFGGDDSGPGGFLNINSLP	DFRNPYGFPGSPQAPLG	YLDDESM	LPPQDFD	DFDSGQFVFP	PGYPGSPGSGSQFP	188	
Lcorn 58650 AgSp1.2 cr	MVGPIGFFPNFNPEIGGQFP	GFGGDDSGPGGFLNINSLP	DFMNPYGFPGSPQAPLG	YLDDESM	LPPQDFD	DFDSGQFVFP	PGYPGSPGSGSQFP	188	
Ncruc 58532and13532 AgSp1 cr	MVGNIGFFPNFSPEIGGPF	GFGGDDSGPGGFLNIGSLP	NFKSPGYGFPGSPQAPLG	YLDQNM	LPPGYNP	DFSGPLVFP	PGYSGSPGSGGQLPG	190	
Mlaby 7048 AgSp1 cr	QVGSIGYFPDFSGEIGGPF	GFGSGGPGGPGGFLNIMSLP	DFNRGYGFPGSPQAPLG	YLDDESM	LPPKGFNP	NFSGQFVYP	PGYPGSPGSGGQYPG	188	
Mhut 1271 AgSp1 cr	MIGSIGFLPNFSPEIGGPF	GFGGDDSGPGGYLNIDSLP	DFNKPYGFPGSPQAPLG	YLDKSK	LPPGYNP	GFSGQLVFP	PGYPGAPGSGSQFP	188	
Aarg AgSp1 FC511 cr	MEGSIGFLPNFSPEIGGPF	GFGGPGSPGLSGFLNINSLP	DFINQYGFPGSPQDPLG	YLNFS	M	LPPGYNP	DFSGQFVYP	PGYPGSPGSGGQFP	188
Atri AgSp1 genomic cr	MEGPPIGFLPNFSPEIGGPF	GFGGPGSSGPGGFLNINSLP	DFINQYGFPGSPQDPLG	YLNFS	M	LPPGYSP	DFSGQFVYP	PGYPGSPGSGGQFP	188
Msag 19329 AgSp1.1 cr	IAPPGISPNFGLEIGGVFP	SISAGADDSAPGEHLNIH	STPNLINPLYGFPSFQAP	LGFYSSQ.....	SS	EFSDKFVL	PGDLLSPGSEGGQLG	108	
Msag 18528 AgSp1.2 cr	FDGSIGFFPNFNPEIGGPF	GFGGEDSGPGGYLDIDSLP	DFINPLYGFPGSPQAPLG	YIHNHP.....	SSDVS	DTFVLPGD	PAPGSGQFP	181	
Mg TRINITY DN1546 c0 g1 Cr	MDDSVGHFNPFLNLEIGGS	FLGYEGGPNNLGLGENLN	ITLPIFYIKPGSGFGSPG	SHVPFG	HFHDYNQL	LPPDFNP	NFPGLFALPGYSGFP	ESRSQFAG	142
Varen 6869 AgSp1.1 cr		SFRPSG	GDPSNLGPDGSLNTDS	IPDFDNP	GYGSPGYQVPLGY	FD	SVP.....	NSDPSSQFVLPGSPGSDGQFP	109
Varen 6869 AgSp1.2 cr	MYGSIGFLPNFSPEIGGPF	GFGGDFGPGGYLNTDSLP	DFINPGYGFPGSPQAPLG	YDFN	KLPPDFNS	DFSGM	FVLPGSPGSPGSGNQFP	182	

Amarm 50219 AgSp1.1 cr	GFLSFPQLPKIITDKIDGTFS	SFPQLIEALQPLFPGQNI	NIGAIPKDKLQDIPG	FE	CTYDNLKLANFG	GNNQPTGGVFYLPE	MVRLIGYLPVGSF	283	
Lcorn 51676 AgSp1.1Cfrag cr	GFLSFPQLPKDITDKINGTFS	SFPQLINALQPLFPGQNI	NIGAIPQDKLQNI	IPGFNG	SYDNLKLANFG	DNSQPTGGVFYLP	ELVRLISYLPVGSF	283	
Lcorn 58650 AgSp1.2 cr	GFLSFPQLPKDITDKISGTFS	SFPQLINALQPLFPGQNI	NIGAIPQDKLQNI	IPGFNG	SYDNLKLANFG	DNSQPTGGVFYLP	ELVRLISYLPVGSF	283	
Ncruc 58532and13532 AgSp1 cr	GFLSFPQLPKDITDKISSTFS	PELIKSLQPLFPGQNI	MGAIPQDKLQNI	IPGFNG	TYDNLKLAN	YDNGQPTGGVFYLP	ELVRLMSYLPVGSF	285	
Mlaby 7048 AgSp1 cr	GFLSFSQLPESITNIHGS	SFSFSQMMNALG	PLFPQQVNIGAIPK	NQLQS	IPGFNG	NYDNLMLDN	YGDSSQPTGGVFYLP	ELVRLASYP	282
Mhut 1271 AgSp1 cr	GFLSFPQLPKDITDKISGTFS	SFPQLINSLQPLFPGQNI	GINIGTIPQDKLQD	IPGFTGTYN	NLKLANFG	DNSQPTGGVFYLP	ELVRLIGYLPV	283	
Aarg AgSp1 FC511 cr	GFMSFPELPKDVTDKINGT	FSLPQLMQALQPLFPGQNI	GINIGAIPKQMQN	IPGFEG	TYDNLRLANFG	NNQPVGGVFYLP	ELVRLISYLPVGSF	283	
Atri AgSp1 genomic cr	GFLSFSSELPQGVTDKINGT	FSLPQLMQALQPLFPGQNI	GINIGAIPKQMQN	IPGFKG	TYDNLRLANFG	SDNQPTGGVFYLP	ELVRLTSYLPVGSF	283	
Msag 19329 AgSp1.1 cr	GFLSFPQLSKDITNLING	SFSFAELIQALQPLFPG	NSINVGAI	PHSELEV	MPGFNG	TYDNLKLDNI	KNNGH	PNGGVFYLP	203
Msag 18528 AgSp1.2 cr	GFLSFPQLPKSITDMINGT	FSFQELIQALQPLFPG	NSINVGAI	PD	DKLQDIPGL	KGNHDSLKL	SQLGDNSQPSGSLFY	IPELMILISHLPD....	272
Mg TRINITY DN1546 c0 g1 Cr	GFLSFPQLPKGITNNING	SFSFAELIQALQPLFPG	KSINVGAI	PNKLQV	MPGFNG	TYDNL	ELGDI	GDSSHPN	237
Varen 6869 AgSp1.1 cr	GFLSFMQLPKSITDKINRT	FTFELIQAVQPLFPGNT	INLGAIP	RDNLQDIPGF	DGSYNNLK	FDYFGDSTKQ	NGGIFYIPEL	IYLINQLPG....	200
Varen 6869 AgSp1.2 cr	GFLSFPQLPKDITDKIDR	PFSFPQLIAIQPLFPG	DSINVGAI	PE	DKLKDMPGF	NGTSDSLRL	DNFEDNNR	PSGGVFYIPEL	277

Amarm 50219 AgSp1.1 cr	YGPGTINHDGGFGHPFNFPGLNGAPGYICDYPNNG..VL...SQDLGGEVKGNNGNGPSGDVADAAPENDLGGTAPENDIGGAAP.....N.	364
Lcorn 51676 AgSp1.1Cfrag cr	KSGGTINQDGGFGDPFNFPGLNGAPGYICDYPDNG..DATAGGSQDLGGEVKGNNGNGPSGDVADAAP..DLGSPAPENDLGGAAAP.....	364
Lcorn 58650 AgSp1.2 cr	KSGGTINQDGGFGDPFNFPGLNGAPGYICDYPDNG..DATAGGSQDLGGEVKGNNGNGPSGDVADAAP..DLGSPAPENDLGGAAAP.....	364
Ncruc 58532and13532 AgSp1 cr	KGPGTINHDGGFDDPFNFPGLNGAPGSICDYPDNG..DATDGGSQDLGGEVKDGNGPAGDVADAAP..DLGAPAPGNDLGGAAAPDLGAPAPGND	376
Mlaby 7048 AgSp1 cr	ANGGFGDPFGEQLLHGAPGYICDYPDND..DTTAGGSQDLGGEIKGNHGPSGDVADAAP..NLGGSAPGNDLGGAAAPG.....NT	359
Mhut 1271 AgSp1 cr	KDSEIINKNGGFSNPFNFPGLNGFPGYICDYPDNG..DATTEGSQDLGGEIEGYGNPFPEDVADAAPENDLE.....	353
Aarg AgSp1 FC511 cr	NGPGTINQDGGFGDPFNFPGLNGAPGYICDYPDNG..DAATGRSQNLGGEIKGNNGNGPSGDVADAAPADLGGAAAPQSDLA.....	362
Atri AgSp1 genomic cr	NGPGTINQNGGFGDPFNFPGLNGAPGYICDYPDNG..HATTGGSQNLGGEIKGNDNGPSGDVADAAPGNNLGGAAPGNNLGSAAPG.....N.	368
Msag 19329 AgSp1.1 cr	KGLGTINQEGGFGHPVNFPGLSGTLGYICDYLENG..DATIEGSQIFS.....	249
Msag 18528 AgSp1.2 cr	IGPETINSDGGFSTPFNFPGFSGPFGYICDYPDKD..DVTDCGFHDLGGGVKGNDNRPFGDVCGAAPDEDDANDVASQDLGGEVK..INKTGPFRD	364
Mg TRINITY DN1546 c0 g1 Cr	IGPGTINQEGGFGDAIKFPGLNGAPGYICDYPENG..DATDRGSQDLGEGVFGDGNTDTNDDGFQES.....	302
Varen 6869 AgSp1.1 cr	GISGTDFDFPGFYGPRRNICDAGN...DSSDSGSQDLGGEVEGDLGLFGDMGDAAPSDTMDAAPR.....	263
Varen 6869 AgSp1.2 cr	KSGGTINQDGGFGEFPFNFPGLNGAPGNICDYPEDYYDYSMDEGSQDLGGEVEGDGNGPSGDVGDAAAPEDIMGAAPPNLSP.....	358

Amarm 50219 AgSp1.1 cr	DLGGA.....PENDLGGAAAP.....ENDLGGAAAPQSEMG..SNTLVA	400
Lcorn 51676 AgSp1.1Cfrag cr	DLGSPA.....PENDLGGVAP.....NLGSPAPQTNQG..SNTLVA	398
Lcorn 58650 AgSp1.2 cr	DLGSPA.....PENDLGGVAP.....NLGSPASETNQG..SNTLVA	398
Ncruc 58532and13532 AgSp1 cr	LGGAAPDLGAPAPG..NDLGGAAPDLGAPAPGNLGGAAAPDLGAPAP..GNDLGGAAADLG.....APAPAPGNDLGATDPPSDQ..GSNTLVA	459
Mlaby 7048 AgSp1 cr	LGGSA.....TLGGSAP.....GNAPGGSAPGNM..ASNTLVG	394
Mhut 1271 AgSp1 cr	GVASQSNLG..PNTLVA	368
Aarg AgSp1 FC511 cr	SPTPVA	368
Atri AgSp1 genomic cr	NLGSAA.....PGNNLGSAAAP.....ENDLGGAAAPENNAPQSDTPSS	406
Msag 19329 AgSp1.1 cr		249
Msag 18528 AgSp1.2 cr	VGGAAPNEDGANDGGSQDLGGEV..KSNKVGPFVRDVGGAAAPYEDGANDGDSQDLGGEVKGSGNGPSGDVADAAPDDNMADAPQSNL..SSPTPVT	456
Mg TRINITY DN1546 c0 g1 Cr	GKEVKDNRNGSSGDVGNSFPDYSIVGVTPLTNQS.....SGA	339
Varen 6869 AgSp1.1 cr		263
Varen 6869 AgSp1.2 cr	P	359

Amarm 50219 AgSp1.1 cr	PEDEDCQDNVPEAFNKKARQAAT	422
Lcorn 51676 AgSp1.1Cfrag cr	PDDEDCQDNVPETFSKAREALA	420
Lcorn 58650 AgSp1.2 cr	PDNEDCQDNVPETFSKAREALA	420
Ncruc 58532and13532 AgSp1 cr	PDDEDCQDNVPEAFHKKARQALA	481
Mlaby 7048 AgSp1 cr	PEDEGCQDNVPEAFSKARQALA	416
Mhut 1271 AgSp1 cr	PEVKDCQDSVPEAFRKARHALT	390
Aarg AgSp1 FC511 cr	PTDQDCCEEDVHGAFSKARQALM	390
Atri AgSp1 genomic cr	PTDKGCEEDVQGAFSKARQALM	428
Msag 19329 AgSp1.1 cr		249
Msag 18528 AgSp1.2 cr	GTEGNCEDVDRTLNTAKQALQ	478
Mg TRINITY DN1546 c0 g1 Cr	DIDGNCNDNVATTFNNTTKQAIR	361
Varen 6869 AgSp1.1 cr	..NDPSCGDDVNSAFSTAKQALQ	284
Varen 6869 AgSp1.2 cr	TSVTLCEDNVVSTFSAKQALQ	381

☐ non-conserved
☒ similar
☒ ≥ 50% conserved